A MID-TERM PROGRESS REPORT ON

AI-Enabled Hydrological Information System for the Mahanadi Basin

Prithvi Flood-IN: Pre- and During-Event Inundation Mapping with Prithvi-EO-2.0 over the Indian Subcontinent

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Abstract

Rapid, reliable maps of flooded area are essential for flood response in India, yet official maps are produced largely by expert interpretation and physics-based models are limited by costly, uncertain river-geometry and elevation data. Working with ClimMaTech Pvt. Ltd. towards an AI-enabled hydrological information system for the Mahanadi basin, this project adapts a geospatial foundation model to Indian floods and links it to physically based flood routing. We curated an India-specific dataset by pairing Sentinel-2 six-band imagery with NDEM flood-extent maps: screening 1,022 NDEM layers yielded 180 scene–label pairs from 100 events (2021–2026) and 3,584 chips of 512×512 pixels, with clouds, shadows and permanent water set to *ignore* and an event-wise 72/12/16 train/validation/test split. Full fine-tuning of the 300 M-parameter `Prithvi-EO-2.0-300M-TL-Sen1Floods11` model with TerraTorch on a free T4 GPU raised flood-class IoU on 16 unseen events from 0.337 to 0.375 (+11%) and precision from 0.391 to 0.560 (+43%), while recall fell from 0.711 to 0.532. A mentor-provided benchmark on Sen1Floods11 showed TerraMind-small (IoU 0.807) ahead of a U-Net (0.765), a simple CNN (0.762) and Otsu thresholding (0.230). We also implemented a MacCormack Saint-Venant solver, quantified the sensitivity of routed peaks to channel parameters, studied a PINN code base, designed an inference and decision-support workflow, and visited Pawna Dam. Next steps include threshold calibration, Sentinel-1 SAR fusion, FastFlood benchmarking, PINN validation and a Mahanadi pilot.

Keywords: flood inundation mapping; foundation model; Prithvi-EO-2.0; Sentinel-2; NDEM; Saint-Venant equations; PINN; Mahanadi basin.

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1 Introduction

1.1 Background

Floods are the most frequent and most damaging natural hazard in India. Every south-west monsoon (June–September) brings riverine flooding to the Ganga, Brahmaputra, Mahanadi, Godavari and Krishna basins, while short-duration extreme rainfall causes flash and urban floods in hilly and densely built regions. The consequences — loss of life, displacement, damage to crops, roads, bridges and power infrastructure — recur annually, and climate change is expected to intensify extreme rainfall.

Effective response during a flood depends on knowing *where the water is*. Disaster-management authorities need inundation extents within hours of a satellite overpass to prioritise evacuation, route relief, identify cut-off villages and estimate crop and infrastructure losses. In India, official flood layers are disseminated by the National Database for Emergency Management (NDEM) of ISRO/NRSC. These layers are of high quality but are produced through analyst-supervised interpretation, which limits how quickly and how often they can be generated.

Two complementary families of tools are available to automate this task:

1. **Physics-based hydrodynamic models**, which solve the Saint-Venant (shallow-water) equations to predict how a flood wave propagates through a river network and onto the floodplain. Their accuracy is limited by the availability of river cross-sections, bed slope, roughness and elevation data.
2. **Data-driven Earth-observation (EO) models**, which classify satellite imagery pixel by pixel into flooded and non-flooded land. Recent *geospatial foundation models*, pre-trained on millions of unlabelled satellite images, promise strong performance from comparatively few labelled examples.

This project brings the two together in the context of an industry product.

1.2 Industry context: ClimMaTech and the INCLINE/LSHMS platform

ClimMaTech Pvt. Ltd., an IIT Mandi-incubated company, together with Prof. Basudev Biswal's group at IIT Bombay, is developing a hydrological information and flood-modelling platform (referred to internally as INCLINE/LSHMS). At the project kick-off (3 August 2026) the team was introduced to the existing software, and the following characteristics and gaps were identified:

- The platform uses two families of input: *topographic data* (river geometry and elevation) and *rainfall data* (observed records for two pilot regions, and future rainfall for identifying flood-prone areas).
- Field surveys of river geometry are expensive — of the order of **Rs. 1 lakh for 10–15 km** of river — so river width is taken from observations while depth is estimated through empirical relations.
- Available elevation data carry a vertical uncertainty of about $\pm$**1 m**, and slopes are derived through a smoothing algorithm that offers scope for improvement.
- The platform already generates automated, LLM-written flood reports in 16 languages.

A satellite-derived, observed flood extent is therefore valuable to the platform in two ways: directly, as an inundation product for decision support, and indirectly, as an independent observation against which physically based simulations can be checked and calibrated.

1.3 Problem statement

Prithvi Flood-IN: Pre- and During-Event Inundation Mapping with Prithvi-EO-2.0 over the Indian Subcontinent.

The publicly released flood-segmentation version of IBM–NASA's Prithvi-EO-2.0 foundation model was fine-tuned on the global Sen1Floods11 dataset, which contains only 68 Indian chips out of 446. Indian flood scenes differ systematically from that training distribution: rivers carry high sediment loads, monsoon cloud cover is persistent, and land cover ranges from irrigated paddy to dense urban areas. Applied to Indian events, the published model detects most flooded area but produces many false alarms (Chapter 6). The problem addressed is to *adapt* this foundation model to Indian conditions using official Indian flood maps, to evaluate it rigorously on unseen events, and to define how its output can be integrated with hydrological data and physics-based reasoning for flood response in the Mahanadi basin.

1.4 Objectives

The aim of the project is to contribute towards the product development of the ClimMaTech–IIT Bombay team. The specific objectives are:

- **O1.** Develop flood inundation mapping with machine-learning techniques using available satellite images.
- **O2.** Build an India-specific flood training dataset by pairing Sentinel-2 imagery with NDEM flood-extent maps from real Indian flood events, with rigorous quality control.
- **O3.** Fine-tune the Prithvi-EO-2.0 foundation model for Indian conditions and quantify its improvement over the published model on unseen events.
- **O4.** Benchmark foundation models against classical and from-scratch deep-learning baselines.
- **O5.** Develop an inference workflow that produces georeferenced flood-extent maps and design their integration with dam, gauge and infrastructure layers.
- **O6.** Study physics-based flood routing through the Saint-Venant equations and physics-informed neural networks, and link satellite observations to physics-based model validation.
- **O7.** Pilot the system on the Mahanadi basin and prototype a decision-support dashboard.

1.5 Scope of the mid-term report

This report documents work completed between 29 July and 5 October 2026: the literature and platform study, dataset curation, baseline evaluation and full fine-tuning of Prithvi-EO-2.0, the benchmark study on Sen1Floods11, the Saint-Venant and PINN study, the inference and application design, and a field visit. Objectives O1–O4 are substantially complete; O5 and O6 are in progress; O7 is planned for the second half of the semester.

2 Literature Review

2.1 Remote sensing of floods

Optical sensors such as the Multispectral Instrument on Sentinel-2 measure reflected sunlight in 13 spectral bands at 10–60 m resolution with a revisit of about five days. Open water absorbs strongly in the

near-infrared (NIR) and shortwave-infrared (SWIR) bands, so spectral indices such as the Normalised Difference Water Index, $\mathrm{NDWI} = (G - \mathrm{NIR})/(G + \mathrm{NIR})$, and its modified form using SWIR, $\mathrm{MNDWI} = (G - \mathrm{SWIR1})/(G + \mathrm{SWIR1})$, have been used for decades to delineate water. The principal limitation of optical sensing during floods is cloud cover, which is especially persistent during the Indian monsoon.

Synthetic aperture radar (SAR), such as the C-band Sentinel-1, is an active sensor that penetrates cloud and works day and night. Calm water acts as a specular reflector and appears dark (typically below about -15 dB in VV and -22 dB in VH polarisation). SAR has well-known failure modes: smooth dry surfaces (tarmac, sand) and radar shadow are also dark, producing false alarms; flooded vegetation produces bright double-bounce returns, producing misses; wind roughens water surfaces; and urban areas scatter strongly. Optical and radar data therefore fail in different situations, which motivates multi-sensor approaches.

2.2 Classical thresholding

The most widely used operational method for SAR flood mapping is histogram thresholding, in particular Otsu's method [7], which chooses the grey-level threshold that maximises the between-class variance of a bimodal histogram. It needs no training data, but its performance degrades when water occupies a small fraction of the scene (the histogram is not bimodal) or when land and water backscatter overlap.

2.3 Deep learning for flood segmentation

Convolutional neural networks (CNNs) learn spatial and spectral features directly from labelled data. The U-Net [8] — an encoder–decoder with skip connections — has been the default architecture for semantic segmentation since 2015 and remains a strong baseline. The Sen1Floods11 dataset [1] provides 446 hand-labelled 512×512 chips at 10 m resolution from 11 flood events across six continents, with co-registered Sentinel-1, Sentinel-2, DEM and JRC permanent-water layers, and has become the standard benchmark for flood segmentation. Its official splits share events between training and test sets, so reported scores measure generalisation *within* seen events rather than transfer to new floods.

2.4 Geospatial foundation models

A foundation model is a large network pre-trained on a very large unlabelled corpus with a self-supervised objective, typically reconstructing masked portions of its input, and then adapted to downstream tasks with relatively few labels.

- **Prithvi** [3] was released by IBM and NASA as a Vision Transformer (ViT) masked autoencoder pre-trained on Harmonized Landsat–Sentinel-2 (HLS) imagery over the United States.
- **Prithvi-EO-2.0** [2] extends pre-training to a global, multi-temporal HLS corpus and releases 300 M and 600 M parameter variants, together with task-specific checkpoints including `Prithvi-EO-2.0-300M-TL-Sen1Floods11`, which is the starting point of this project. It uses six HLS bands: Blue, Green, Red, narrow NIR, SWIR-1 and SWIR-2.
- **TerraMind** [4] is a multimodal, generative foundation model pre-trained jointly on Sentinel-2 L1C/L2A, Sentinel-1 GRD/RTC, DEM, NDVI, land cover and RGB, released in tiny (6.1 M), small (22.8 M), base (87.9 M) and large (305.9 M) sizes.
- **TerraTorch** [5] is IBM's open-source fine-tuning toolkit built on PyTorch Lightning, which attaches task heads to pre-trained encoders, handles multiband GeoTIFF loading and normalisation, and expresses complete training runs as reproducible YAML configurations.

Whether foundation models actually outperform well-tuned supervised baselines is contested: the PANGAEA benchmark [6] evaluated twelve geospatial foundation models and found that a plain supervised U-Net outperformed all of them on Sen1Floods11. Results are also highly sensitive to hyper-parameters such as the learning rate [14]. These findings motivated the benchmark study in Chapter 5.

2.5 Physics-based flood routing

Unsteady open-channel flow is governed by the one-dimensional Saint-Venant equations [11], expressing conservation of mass and momentum. Depending on which momentum terms are retained, routing models are classified as kinematic-wave, diffusive-wave or full dynamic-wave models. Operational hydrodynamic packages such as HEC-RAS and LISFLOOD-FP solve these equations in one or two dimensions to predict stage, discharge and inundation extent. Their accuracy depends on channel geometry, roughness and boundary conditions — precisely the inputs identified as costly and uncertain for the ClimMaTech platform.

2.6 Physics-informed neural networks

Physics-informed neural networks (PINNs) [12] train a neural network to approximate the solution of a PDE by minimising a loss that combines misfit to observations with the residual of the governing equations, evaluated through automatic differentiation at collocation points. For river hydraulics, PINNs can assimilate sparse gauge observations while respecting the Saint-Venant equations, offering a route to estimating flow states and uncertain parameters between gauges.

2.7 Research gap

From the review, three gaps directly relevant to this project emerge:

1. Published foundation-model flood checkpoints are trained and evaluated predominantly on non-Indian data, with evaluation splits that share events between training and testing.
2. There is no openly documented, India-specific, event-wise-split training set derived from official NDEM flood maps.
3. Satellite-observed flood extents and physics-based routing are rarely combined within a single operational information system for Indian river basins.

This project addresses the first two gaps in its first half and is designed to address the third in its second half.

3 Study Region and Data

3.1 Study region

The geographic scope of the dataset is the Indian subcontinent, using every NDEM flood layer for which matching satellite imagery could be found. The application pilot is the **Mahanadi basin**, one of India's major east-flowing river systems, draining parts of Chhattisgarh and Odisha into the Bay of Bengal. The basin is regulated by the Hirakud reservoir and is prone to monsoon floods in its deltaic reaches, making it a natural test-bed for combining reservoir operations, river gauging and satellite flood mapping.

3.2 Sentinel-2 multispectral imagery

Input imagery comes from the Copernicus Sentinel-2 mission (ESA), which is freely available. Six bands, in the order expected by Prithvi-EO-2.0, are used (Table 3.1). The two SWIR bands carry the strongest water signal because water absorbs strongly at these wavelengths.

Table 3.1: Sentinel-2 bands used as model input, in channel order.

Ch.	Band	Sentinel-2 band	Central wavelength (nm)	Role in flood mapping
1	Blue	B02	490	Turbidity, atmospheric effects
2	Green	B03	560	Water reflectance peak; NDWI
3	Red	B04	665	Vegetation and sediment contrast
4	NIR (narrow)	B8A	865	Strong water absorption; vegetation
5	SWIR-1	B11	1610	Very strong water absorption; MNDWI
6	SWIR-2	B12	2190	Very strong water absorption; burn/soil

3.3 NDEM flood-extent maps (ground truth)

Ground-truth labels are the official flood-inundation layers published by NDEM (ISRO/NRSC) for real Indian flood events from 2021 to 2026. NDEM layers mark *newly inundated* land for a given date; they do not mark permanent rivers and lakes as flooded. This property matters for label design (Section 4.2.2).

3.4 Sen1Floods11 (benchmark dataset)

For the benchmark study, the global Sen1Floods11 v1.1 hand-labelled subset [1] was used (1.8 GB). It contains co-registered Sentinel-1 GRD (VV, VH), Sentinel-2 L1C (13 bands), hand labels (1 water, 0 dry, -1 undetermined), Otsu labels, JRC permanent water and Copernicus DEM for 446 chips of 512×512 pixels.

The official split has 252 training, 89 validation and 90 test chips from ten countries; India contributes 68 chips (40/14/14), and all ten countries appear in more than one split, so the benchmark measures generalisation within seen events.

An exploratory scan of 431 label files showed a mean water fraction of 0.107 per chip, 145 chips with less than 1% water, 26 chips with more than 50% water, and a mean unlabelled (-1) fraction of 0.132. Class balance therefore varies enormously from chip to chip, and a significant share of pixels must be excluded from both the loss and the metrics. In sample chips, median VH backscatter was -19.1 dB over water and -15.3 dB over land, with 50.3% of water pixels brighter than the darkest 5% of land pixels, so a fixed threshold cannot separate the classes.

4 Methodology

4.1 Overall workflow

The methodology has four connected components: (i) construction of an India-specific labelled dataset; (ii) adaptation of the Prithvi-EO-2.0 foundation model by full fine-tuning; (iii) fair, leakage-free evaluation against the published model; and (iv) an inference workflow that converts model output into decision-support products. A parallel physics track (Chapter 7) studies flood routing.

4.2 Dataset construction

4.2.1 Pairing satellite scenes with flood maps

For each NDEM flood layer, a Sentinel-2 scene was sought that satisfied three conditions: (a) acquisition *during* the mapped flood, close to the NDEM map date; (b) spatial overlap with the mapped flood area; and (c) sufficiently low cloud cover over the flooded region. An automated pipeline searched candidate scenes, checked cloud cover, verified that the flood outline overlapped visibly wet pixels, and selected the best scene per event. A sample of matches was further verified visually by overlaying the NDEM outline on the false-colour image.

The initial project spreadsheet listed 221 flood events, of which only 5 were directly usable. Extending the search to all 1,022 dated flood-extent layers on the NDEM server yielded **180 usable scene–label pairs covering 100 distinct flood events** (Figure 4.1). The main reasons for rejection were absence of imagery on suitable dates, cloud cover over the flooded area, and insufficient overlap between the scene and the mapped flood.

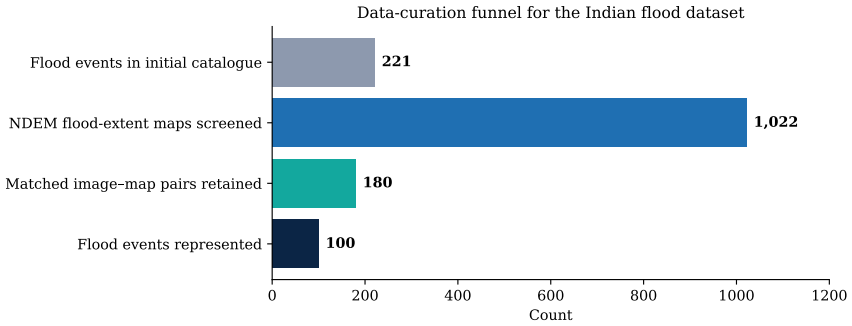


Figure 4.1: Data-curation funnel from the initial event catalogue to the final dataset.

4.2.2 Label design

Each pixel of the target raster takes one of three values:

- **1 — flood**: inside the NDEM inundation polygon;
- **0 — dry land**: inside the scene and outside the inundation polygon;
- -1 **— ignore**: cloud, cloud shadow, permanent water (rivers, lakes, reservoirs) or area outside the officially mapped flood zone.

Assigning permanent water to the ignore class is essential. Because NDEM marks only newly flooded land, labelling the river channel as "dry" would teach the model that open water is dry land. Similarly, cloud-covered pixels cannot be verified and are excluded rather than guessed. The ignore value is passed to the loss as `ignore_index = -1`, and the same mask is applied when computing every metric. The construction pipeline is summarised in Figure 4.2.

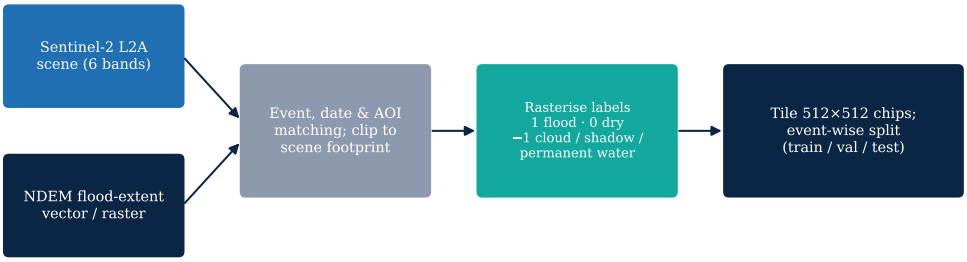


Figure 4.2: Pipeline for constructing training chips from Sentinel-2 scenes and NDEM flood maps.

4.2.3 Tiling and storage optimisation

Scenes and label rasters were tiled into chips of 512×512 pixels. This gave **3,584 chips**, of which 2,401 contain flooded pixels and 1,183 do not. Several NDEM files were very large (over 150 MB) because they covered regions far bigger than the matching scene; clipping each label to its scene footprint reduced the dataset from about **1.9 GB to 625 MB**.

4.2.4 Event-wise data split

Two scenes from the same flood are highly correlated. If one were used for training and the other for testing, the test score would reward recognising the scene rather than understanding floods. The split was therefore performed by *whole flood event* (Table 4.1). This is a stricter protocol than the official Sen1Floods11 split, in which every country appears in more than one subset.

Table 4.1: Event-wise split of the Prithvi Flood-IN dataset.

Split	Events	Scenes	Chips	Purpose
Train	72	130	2,635	Weight updates
Validation	12	18	212	Model selection after every epoch
Test	16	32	737	Held out; used once for final scoring
Total	**100**	**180**	**3,584**	

4.3 Model architecture

We reuse the exact architecture and starting weights of the published checkpoint `ibm-nasa-geospatial/Prithvi-EO-2.0-300M-TL-Sen1Floods11`, built with TerraTorch's `EncoderDecoderFactory` inside a `SemanticSegmentationTask` (Figure 4.3).

- **Encoder** (`prithvi_eo_v2_300_tl`): a ViT-Large encoder of about 300 M parameters (24 transformer blocks) pre-trained as a masked autoencoder on global HLS data, already transfer-learned once on Sen1Floods11. The image is divided into patches, each embedded as a token; self-attention models relations between all patches.
- **Neck**: (1) `SelectIndices [5, 11, 17, 23]` taps four evenly spaced encoder depths, combining shallow detail and deep semantics; (2) `ReshapeTokensToImage` folds each token sequence back into a 2-D feature grid; (3) `LearnedInterpolateToPyramidal` learns upsampling to four spatial scales, producing a feature pyramid.

- **Decoder**: `UperNetDecoder` with 256 channels, which fuses the multi-scale features (pyramid pooling and feature pyramid) into a single full-resolution map.
- **Head**: a per-pixel classifier producing two logits (not-flood, flood), with dropout 0.1.

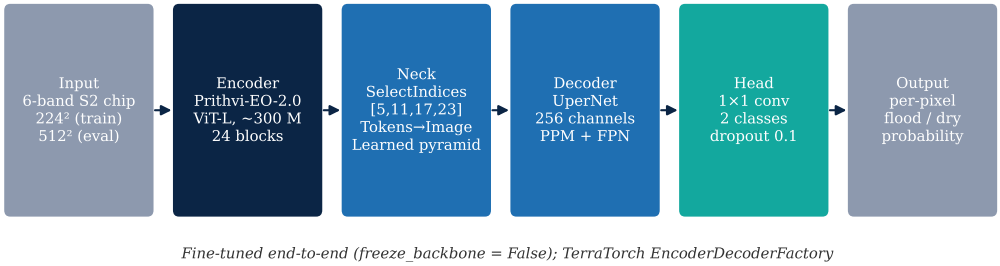


Figure 4.3: Encoder–neck–decoder–head architecture of Prithvi Flood-IN.

4.4 Training configuration

Training uses the per-pixel cross-entropy loss over valid pixels,

$$\mathscr{L} = -\frac{1}{|\Omega|} \sum_{i \in \Omega} \sum_{c \in \{0,1\}} y_{i,c} \log \hat{p}_{i,c}, \qquad \Omega = \{\, i : y_i \neq -1 \,\}, \tag{4.1}$$

where $\hat{p}_{i,c}$ is the softmax probability of class c at pixel i. Table 4.2 lists all settings.

Table 4.2: Training configuration for Prithvi Flood-IN.

Setting	Value	Rationale
Loss	Cross-entropy	Standard per-pixel classification loss
`ignore_index`	-1	Cloud, shadow, permanent water and unmapped pixels excluded
Optimiser	AdamW	Adaptive optimiser with decoupled weight decay
Learning rate	5×10^{-5}	Small, to refine rather than overwrite pretrained knowledge
Weight decay	0.05	Regularisation against overfitting
Scheduler	Cosine annealing	Smooth decay of learning rate
`freeze_backbone`	False	Full fine-tune of the 300 M-parameter encoder
`freeze_decoder`	False	Decoder also adapted
Batch size	8	Fits T4 memory with mixed precision
Training crop	224×224 (random)	Memory and speed; full 512×512 at evaluation
Precision	16-bit mixed	Lower memory, faster training
Maximum epochs	20	Planned budget
Early stopping	Patience 8 on val. flood IoU	Prevents overfitting and wasted compute
Checkpoint selection	Best `val/multiclass jaccardindex_flood`	Most generalisable epoch, not the last

4.5 Evaluation methodology

For each test chip, predictions are compared with labels over the valid pixel set Ω. Accumulating true positives (TP), false positives (FP) and false negatives (FN) for the flood class over all 737 test chips, the metrics are

$$\text{Precision} = \frac{TP}{TP+FP}, \qquad \text{Recall} = \frac{TP}{TP+FN}, \tag{4.2}$$

$$\text{IoU} = \frac{TP}{TP+FP+FN}, \qquad F_1 = \frac{2TP}{2TP+FP+FN} = \frac{2PR}{P+R}. \tag{4.3}$$

IoU is the primary metric because, unlike pixel accuracy, it cannot be inflated by predicting "no flood" everywhere in scenes dominated by dry land. The two are related by $F_1 = 2\,\text{IoU}/(1+\text{IoU})$, which we use as a consistency check. Both the published baseline and the fine-tuned model are scored with identical code, identical metric definitions and the identical test set, so the only variable is the model weights. In addition, the following analyses are prepared: confusion matrices; per-event metrics; inference time per chip; and four-panel visual comparisons (Sentinel-2 image, NDEM truth, prediction, error map with hits, false alarms and misses).

4.6 Computing environment

Training ran on Google Colab (free tier) with an NVIDIA Tesla T4 GPU (15,360 MiB; peak use 7.11 GiB) and 12.7 GB of shared RAM, using Python 3.13, `torch` 2.6.0, `torchvision` 0.21.0, `torchaudio` 2.6.0, `terratorch` 0.99.8, `lightning` 2.5.0 and `rasterio` 1.4.3, with the PyTorch Lightning `Trainer`. Google Drive was used for persistence and the local VM disk for active I/O.

5 Benchmark Study: Does a Foundation Model Beat a U-Net?

5.1 Purpose

A foundation model is more expensive to run and harder to configure than a conventional network; it is worth adopting only if it outperforms a well-built baseline. Before committing to Prithvi-EO-2.0 for the Indian dataset, the team studied a benchmark notebook prepared by the industry mentors [14], which compares a classical threshold, two from-scratch CNNs and the TerraMind foundation model on Sen1Floods11 under a single, fair protocol. The results quoted below are those recorded in the notebook (Colab T4 GPU, 15 September 2026).

5.2 Experimental protocol

- **Data:** Sen1Floods11 official splits — 252 training, 89 validation and 90 test chips.
- **Inputs:** 15 channels (13 Sentinel-2 L1C bands followed by Sentinel-1 VV and VH), standardised with TerraMind's pre-training mean and standard deviation for every model; radar no-data values below -80 dB replaced by the VV mean.
- **Metric:** water-class IoU and F1 over labelled pixels only, using a 0.5 probability cut-off, with identical scoring code for all methods.
- **Loss masking:** pixels labelled -1 excluded from the loss through a binary mask (`counts_in_loss`); the notebook stresses that omitting this step inflates every reported score.

5.3 Models compared

Table 5.1: Methods in the benchmark study.

Method	Parameters	Description
Otsu threshold	0	Per-chip Otsu threshold on Sentinel-1 VH; no training
Simple CNN	14,017	Three 3×3 conv layers with batch norm and ReLU; 7×7 receptive field; no downsampling
U-Net	119,409	Two-level encoder–decoder (width 16), max-pooling, transposed-conv upsampling and skip connections
TerraMind-S	≈22.8 M (backbone)	TerraMind v1 small, multimodal input {S2L1C, S1GRD}; necks SelectIndices [2, 5, 8, 11], ReshapeTokensToImage, LearnedInterpolateToPyramidal; U-Net decoder [256, 128, 64, 32]; 2-class head with 10% dropout

The CNN and U-Net were trained for 10 epochs with AdamW (learning rate 10^{-3}, batch size 4) and masked binary cross-entropy. TerraMind was fine-tuned with TerraTorch from a modified version of the official IBM configuration (`terramind_v1_tiny_sen1floods11.yaml`) using `terramind_v1_small` with the backbone unfrozen, 30 epochs, learning rate 10^{-4}, batch size 8, 16-bit mixed precision and seed 42; the checkpoint used for scoring was `epoch=29-step=930.ckpt`.

5.4 Results

Table 5.2: Benchmark results on the Sen1Floods11 test split (90 chips). Source: [14].

Method	Training	Water IoU	Water F1
Otsu threshold (VH)	None	0.230	0.374
Simple CNN	∼2 min, 10 epochs	0.762	0.865
U-Net	∼4 min, 10 epochs	0.765	0.867
TerraMind-S (fine-tuned)	30 epochs	**0.807**	**0.893**

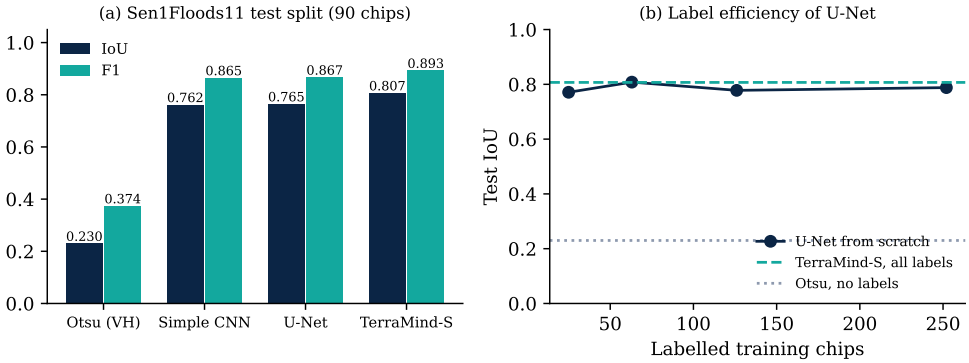


Figure 5.1: (a) IoU and F1 of the four methods on the Sen1Floods11 test split; (b) U-Net test IoU as a function of the number of labelled training chips, with TerraMind and Otsu as reference lines. Data from [14].

5.4.1 Label efficiency

Retraining the U-Net on 10%, 25%, 50% and 100% of the training chips gave IoU values of 0.771 (25 chips), 0.808 (63), 0.778 (126) and 0.788 (252). Going from 25 to 252 chips changed IoU by only +0.017. The curve is essentially flat, which indicates that on Sen1Floods11 the number of labels is not the bottleneck. The 100% value (0.788) differs from the main U-Net run (0.765) despite identical data and seed,

which indicates run-to-run variability of about ± 0.02 IoU on this small test set.

5.4.2 Error analysis

Pixel counts on the chip `Sri-Lanka_713926` (Table 5.3) show that a better IoU does not imply fewer errors of both kinds: TerraMind finds the most water but also raises the most false alarms.

Table 5.3: Error decomposition on chip Sri-Lanka_713926. Source: [14].

Method	False alarms	Missed water	Correct water
U-Net	302	1,644	78,862
Simple CNN	419	1,078	79,428
TerraMind-S	1,339	619	79,887

The notebook also converts a binary map into flooded area, the quantity an emergency manager actually asks for. On chip `Mekong_424793`, the U-Net at a cut-off of 0.5 predicted 0.22 km^2 against a true 0.40 km^2 (IoU 0.474), and the estimated area changed substantially when the cut-off was moved between 0.3 and 0.7. This motivates the threshold-calibration step planned for Prithvi Flood-IN.

5.5 Implications for Prithvi Flood-IN

1. With radar and optical inputs and within-event splits, the foundation model gave a modest but real gain (+0.04 IoU) over a from-scratch U-Net, consistent with the contested picture in the literature [6].
2. Absolute scores on Sen1Floods11 (IoU ≈ 0.8) are not comparable with our event-wise Indian test set, where every test flood is unseen; we expect, and observe, much lower absolute IoU.
3. Masking of unlabelled pixels, fixed-threshold metrics and identical scoring code are non-negotiable for a fair comparison. All three practices are adopted in our evaluation.
4. A from-scratch U-Net baseline on the Indian dataset is a necessary addition in the second half (Chapter 10).

6 Prithvi Flood-IN: Implementation and Results

6.1 Implementation

The project code base is organised into a data module (`floodin/data.py`), a model factory (`floodin/model.py`), a fine-tuning script (`scripts/finetune.py`) and an evaluation script (`scripts/evaluate.py`). The data module reads the six-band chips and label rasters, applies the scaling and per-band standardisation expected by the pre-trained encoder, takes random 224×224 crops during training and full 512×512 chips at evaluation. Training was run from a Colab notebook (`colab_finetune.ipynb`) with the configuration in Table 4.2; checkpoints, logs and result files were stored on Google Drive.

6.2 Engineering challenges and resolutions

Running a 300 M-parameter model on free-tier infrastructure exposed several practical problems. Each was diagnosed by inspecting the running virtual machine (process list, memory use, disk I/O) rather than relying only on the training log, and training was always resumed from the last verified checkpoint.

Table 6.1: Engineering problems encountered and their resolution.

Problem	Symptom / cause	Resolution
Library version mismatch	Pre-installed `torchaudio` incompatible with pinned `torch`; crash on import	Pinned `torchaudio` in `requirements.txt`
Input scaling mismatch	Reflectance not on the scale used in pre-training; baseline model nearly useless	Rescaled inputs to the pre-training convention and applied checkpoint normalisation
Validation plotting crash	Default `plot_on_val` called a dataset interface not implemented	Set `plot_on_val=False`
Orphaned training process	After a Colab disconnect, a stale process kept writing to and corrupting the checkpoint on Drive	Terminated the process; restored last intact checkpoint; write locally first, then sync to Drive
Silent RAM exhaustion	Data-loader workers gradually consumed 12 GB RAM; OS killed training without error	`num_workers=0`, trading speed for stability

6.3 Training outcome

Training ran for 15 of the planned 20 epochs, after which early stopping triggered: no epoch from 8 to 15 improved on the best validation flood IoU of **0.2989**, reached at **epoch 7**. The epoch-7 checkpoint (`run1/checkpoints/best-07.ckpt`) was used for all test results. Peak GPU memory was 7.11 GiB.

6.4 Results on the held-out test set

Table 6.2 and Figure 6.1 compare the published Sen1Floods11 checkpoint (no Indian fine-tuning) with Prithvi Flood-IN on the 737 test chips from 16 unseen events. F1 is computed from precision and recall and agrees with the IoU through $F_1 = 2\,\mathrm{IoU}/(1+\mathrm{IoU})$.

Table 6.2: Flood-class metrics on the held-out test set (16 events, 32 scenes, 737 chips).

Metric	Baseline (published)	Prithvi Flood-IN	Absolute change	Relative change
IoU	0.3372	**0.3752**	+0.0380	**+11.3%**
Precision	0.3909	**0.5601**	+0.1692	**+43.3%**
Recall	**0.7106**	0.5324	−0.1782	−25.1%
F1-score[†]	0.5045	**0.5459**	+0.0414	+8.2%

[†] Derived as $2PR/(P+R)$ from the reported precision and recall.

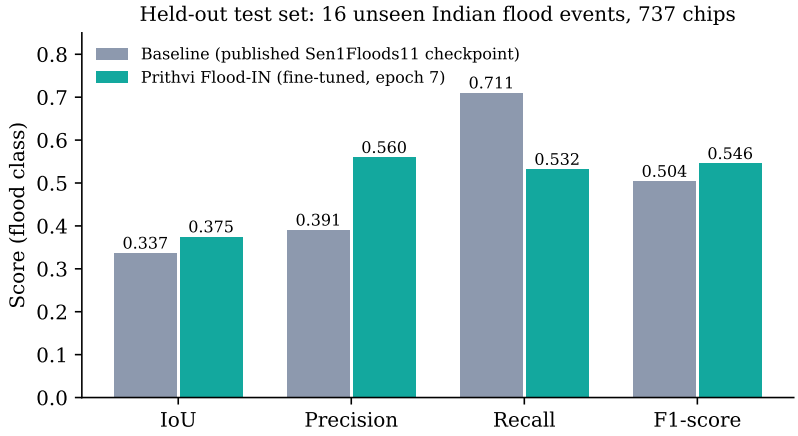


Figure 6.1: Baseline versus fine-tuned metrics on the held-out Indian test set.

6.5 Interpretation

- **The published model was over-eager.** It detected 71% of truly flooded pixels but only 39% of its flood predictions were correct, i.e. roughly three of every five "flood" pixels were false alarms on Indian scenes.
- **Fine-tuning recalibrated the model to Indian conditions.** Precision rose to 56%, cutting the false-alarm share from 61% to 44% of flood predictions.
- **The primary metric improved.** IoU rose by 11.3% and F1 by 8.2%, confirming a net improvement rather than a mere shift in operating point.
- **There is a precision–recall trade-off.** Recall fell from 0.71 to 0.53: the model became more conservative. The fine-tuned model moved to a higher F1 while shifting towards precision. Which operating point is preferable depends on the use: early warning favours recall, whereas damage and insurance assessment favour precision. Because the head outputs a probability, the operating point can be moved by calibrating the decision threshold, which is planned next.
- **Absolute scores are low compared with Sen1Floods11 benchmarks.** This is expected given the stricter event-wise protocol (every test flood is unseen), optical-only input, the coarse and partly generalised nature of NDEM polygons relative to pixel-level hand labels, and the small training set. The validation IoU (0.299) being lower than the test IoU (0.375) also reflects the small number of validation events (12) and inter-event variability.

Per-event metrics for all 16 test events, confusion matrices, inference time and four-panel visual comparisons (Sentinel-2 false colour, NDEM truth, prediction, error map) have been designed and will be reported in the final report.

7 Physics-Based Flood Routing: Saint-Venant Equations and PINNs

7.1 Motivation

At the start of the project the team was asked to study the Saint-Venant equations and their applications, supported by a code base. Physics-based routing predicts how much water arrives where and when; satellite

mapping observes where water actually went. Understanding both is necessary to integrate the observed flood extent into the ClimMaTech platform.

7.2 Governing equations

For one-dimensional unsteady flow in an open channel with lateral inflow q, the Saint-Venant equations in conservative form are

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = q, \tag{7.1}$$

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x}\left(\frac{Q^2}{A}\right) + gA\frac{\partial h}{\partial x} = gA\,(S_0 - S_f), \tag{7.2}$$

with the friction slope given by Manning's equation,

$$S_f = \frac{n^2\,Q\,|Q|}{A^2 R^{4/3}}, \qquad R = \frac{A}{P}, \tag{7.3}$$

where P is the wetted perimeter. Equation (7.1) expresses conservation of mass and Equation (7.2) conservation of momentum (Newton's second law for a control volume), balancing local and convective acceleration, the pressure gradient, gravity and bed friction.

7.2.1 Approximations

Retaining only $S_0 = S_f$ gives the *kinematic-wave* model (steep rivers, no backwater); adding the depth gradient, $S_0 - S_f - \partial h/\partial x = 0$, gives the *diffusive-wave* model, which captures attenuation on mild slopes; the full *dynamic-wave* model is needed for flat reaches, tidal rivers, dam-break and backwater problems.

7.2.2 Applications

The equations underpin flood forecasting and routing (predicting peak magnitude and arrival time downstream), dam-break and reservoir-release analysis, floodplain inundation modelling (e.g. HEC-RAS, LISFLOOD-FP), design of embankments and bridges, irrigation-canal operation, and the calibration of hydrological models against observations — including, in this project, satellite-observed flood extents.

7.3 Numerical solver developed in this project

To build intuition before using the PINN code, a compact explicit solver (`sve_demo.py`, Python/NumPy, about 40 lines) was written for a wide rectangular prismatic channel. In vector form $\partial \mathbf{U}/\partial t + \partial \mathbf{F}/\partial x = \mathbf{S}$ with

$$\mathbf{U} = \begin{bmatrix} A \\ Q \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} Q \\ Q^2/A + \tfrac{1}{2}gBh^2 \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 0 \\ gA(S_0 - S_f) \end{bmatrix}, \tag{7.4}$$

the MacCormack predictor–corrector scheme is

$$\mathbf{U}_i^{*} = \mathbf{U}_i^{n} - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+1}^{n} - \mathbf{F}_i^{n}\right) + \Delta t\,\mathbf{S}_i^{n}, \tag{7.5}$$

$$\mathbf{U}_i^{n+1} = \tfrac{1}{2}\left[\mathbf{U}_i^{n} + \mathbf{U}_i^{*} - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_i^{*} - \mathbf{F}_{i-1}^{*}\right) + \Delta t\,\mathbf{S}_i^{*}\right], \tag{7.6}$$

which is second-order accurate in space and time. The time step is set adaptively by the CFL condition

$$\Delta t = C\,\frac{\Delta x}{\max_i \left(|u_i| + \sqrt{g h_i}\right)}, \qquad C = 0.8, \tag{7.7}$$

which gave $\Delta t \approx 24$–46 s for the conditions simulated.

Table 7.1: Parameters of the reference routing experiment (synthetic channel).

Parameter	Value	Parameter	Value
Reach length L	20 km	Grid spacing Δx	250 m (81 nodes)
Width B	50 m	Bed slope S_0	0.0005
Manning n	0.035	Base flow Q_b	50 $m^3\,s^{-1}$
Normal depth at Q_b	1.31 m	Peak inflow Q_p	400 $m^3\,s^{-1}$ at $t = 3$ h
Hydrograph	Gaussian, $\sigma = 40$ min	Simulated time	10 h
Upstream BC	Inflow hydrograph $Q(0,t)$	Downstream BC	Transmissive (zero gradient)
Initial condition	Uniform flow at Q_b	CFL number	0.8

7.3.1 Results

The simulated hydrographs (Figure 7.1) show the two classical features of flood-wave propagation: *attenuation* of the peak through channel storage and friction, and *translation* (delay) of the peak downstream. The peak fell from 400 to 314 $m^3\,s^{-1}$ at 8 km (arriving 0.76 h later) and to 230 $m^3\,s^{-1}$ at 20 km (2.25 h later), i.e. a mean wave celerity of about 2.5 $m\,s^{-1}$. Part of the attenuation is attributable to numerical diffusion of the scheme, and the outlet node is affected by the downstream boundary condition; the experiment is therefore illustrative rather than a validated simulation.

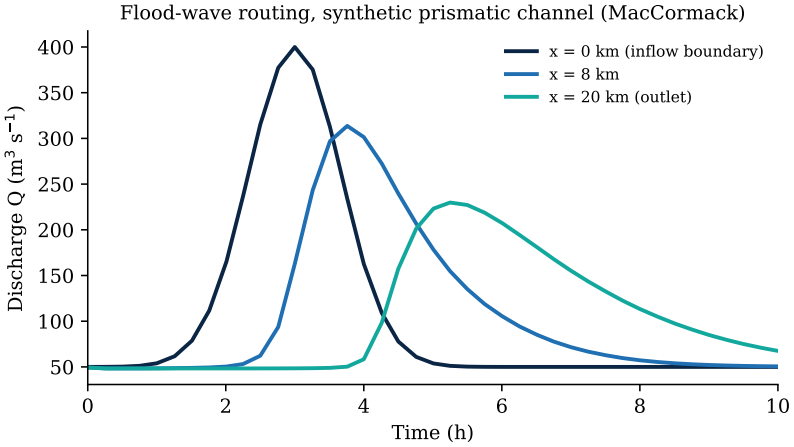


Figure 7.1: Simulated discharge hydrographs at the inflow, 8 km and 20 km.

7.3.2 Parameter sensitivity

Single-parameter perturbations were run to understand how uncertain channel properties — the very quantities that are costly to survey for the platform — affect flood predictions (Table 7.2).

Table 7.2: Sensitivity of peak discharge and travel time at 20 km to channel and flood parameters.

Scenario	Peak at 8 km ($m^3\ s^{-1}$)	Peak at 20 km ($m^3\ s^{-1}$)	Attenuation at 20 km (%)	Travel time to 20 km (h)
Reference	313.7	229.8	42.6	2.25
Rougher bed, $n = 0.050$	276.7	186.1	53.5	3.00
Steeper bed, $S_0 = 0.001$	355.8	303.4	24.2	2.00
Wider channel, $B = 100$ m	298.0	215.2	46.2	3.26
Larger flood, $Q_p = 800\ m^3\ s^{-1}$	643.2	460.0	42.5	1.75

Three observations follow. First, roughness and slope — both uncertain in practice — change the downstream peak by -19% and $+32\%$ respectively, so errors in these inputs propagate directly into flood forecasts. Second, a larger flood travels faster (1.75 h versus 2.25 h), so warning time shrinks precisely when it matters most. Third, wider channels store more water and delay the peak. These results quantify why an independent observation of flood extent is valuable for constraining physically based models.

7.4 Physics-informed neural network code base

The mentors provided a PINN code base for the Saint-Venant equations (`https://github.com/Barbhuiya12/PINN`). A PINN approximates $(A, Q) = \mathcal{N}_\theta(x,t)$ with a fully connected network and trains θ to minimise

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N_d}\sum \left\| Q_\theta - Q^{\text{obs}} \right\|^2}_{\text{data}} + \lambda_r \underbrace{\frac{1}{N_r}\sum \left(r_{\text{mass}}^2 + r_{\text{mom}}^2 \right)}_{\text{PDE residual}} + \lambda_b \underbrace{\frac{1}{N_b}\sum \left\| \mathbf{U}_\theta - \mathbf{U}^{\text{BC/IC}} \right\|^2}_{\text{boundary/initial}}, \tag{7.8}$$

where r_{mass} and r_{mom} are the left-minus-right-hand sides of Equations (7.1) and (7.2), computed by automatic differentiation at collocation points (Figure 7.2).

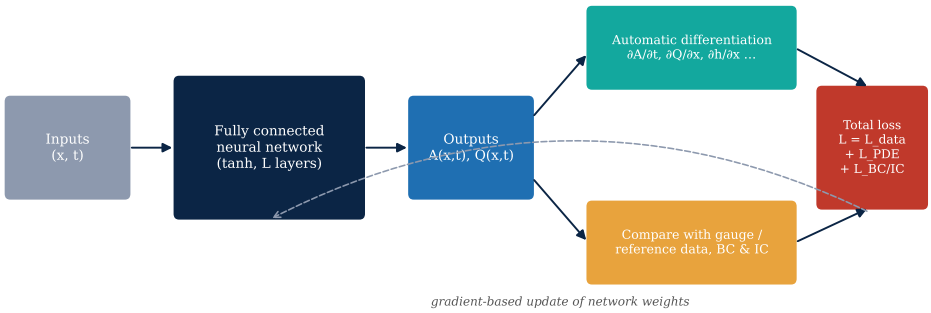


Figure 7.2: Structure of a physics-informed neural network for the Saint-Venant equations.

Table 7.3: Workflow of the mentor-provided PINN code base.

Stage	Description
Setup	`bash SETUP.sh` creates the isolated conda environment `pinn_sve` (Python 3.7, TensorFlow 1.15; not TF2-compatible)
Verification	`python phase4_validation/validate.py -case 3` must report a deviation of 0.0%
Phase 1: data exploration	`p1a_read_reference.py`, `p1b_build_training_sets.py`, `p1c_explore_plots.py` with `-case N`
Training	PINN training on the assigned river reach
Phase 4: validation	Comparison of the PINN solution with the reference solution
Cases	Six cases, each a different river reach (`docs/02_case_assignments.md`)

The code base has been studied together with the Saint-Venant theory above; setup verification, Phase-1 exploration and PINN training on the assigned river reaches are scheduled in the work plan (Chapter 10), where the PINN solution will be compared with the finite-difference solver.

8 Inference Workflow, Applications and Field Visit

8.1 Inference workflow

The deployed model is intended to process a new Sentinel-2 scene acquired during an event and return a georeferenced flood-extent product (Figure 8.1):

1. Acquire the Sentinel-2 L2A scene over the area of interest.
2. Select and order the six bands; apply the scaling and normalisation used in training.
3. Mask cloud, cloud shadow and permanent water.
4. Tile the scene into 512×512 chips and run Prithvi Flood-IN on each chip.
5. Mosaic the per-pixel flood probabilities, apply the calibrated decision threshold, and write a georeferenced raster (GeoTIFF) and vector polygons.
6. Overlay the flood extent on dams and reservoir releases, river-gauge readings, roads and bridges, villages and critical infrastructure (hospitals, shelters, substations) in a decision-support dashboard.

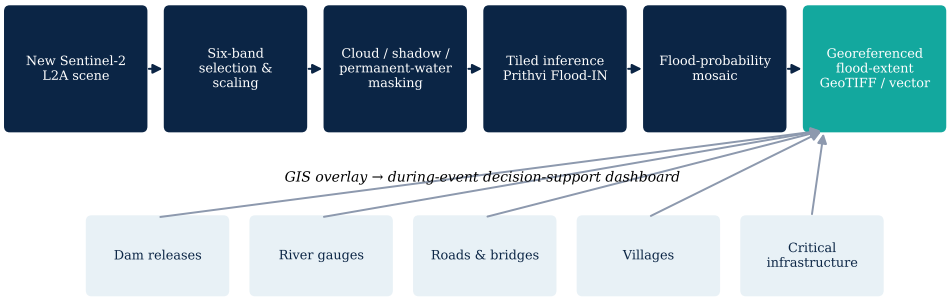


Figure 8.1: Inference and decision-support workflow.

8.2 Applications

Table 8.1: Application pathways for Prithvi Flood-IN.

Application	Description
Mahanadi basin HIS	Near-real-time inundation layer combined with Hirakud reservoir releases and CWC gauge data within the ClimMaTech–IITB platform
Disaster response	Flood-extent maps for NDMA/SDMAs and district control rooms: evacuation planning, relief routing, identification of cut-off villages and roads
Agriculture and insurance	Estimation of inundated cropland per district to accelerate crop-loss assessment and claims under schemes such as PMFBY
Infrastructure risk	Flagging of flooded roads, bridges, substations, hospitals and schools; prioritisation of restoration
Hydrological model validation	Observed extents to calibrate and validate hydrodynamic and rainfall–runoff simulations (Chapter 7)
Flood-risk planning	Multi-year inundation-frequency maps for zoning, embankment planning and climate-risk studies

The precision–recall balance of the model should be chosen per application: higher recall is preferred for early warning and evacuation, where a missed flood is costlier than a false alarm, whereas higher precision is preferred for damage and insurance assessment.

8.3 Field visit: Pawna Dam (13 September 2026)



Figure 8.2: Project team at Pawna Dam, Maval, Pune district, Maharashtra, 13 September 2026.

On 13 September 2026 the team visited Pawna Dam on the Pawna river in Maval taluka, Pune district, at the height of the monsoon. The visit was intended to connect the project's data-driven flood mapping with the physical reality of a regulated, monsoon-fed river system. The main learnings relevant to the project are:

1. **Reservoirs are control points for downstream floods.** Storage in the reservoir attenuates incoming flood peaks, but once the reservoir approaches full supply level, spillway releases pass the flood downstream. Release timing and magnitude therefore largely determine where and when downstream inundation occurs. For the Mahanadi pilot this makes Hirakud release data a core input layer.
2. **Satellite maps need hydrological context.** A flood-extent map shows *where* water is, but not *why* or *what comes next*. Reservoir levels, release schedules and river-gauge readings are needed to interpret an observed extent and to anticipate how it will evolve, which supports the overlay design in Section 8.1.
3. **Observed extents can check routing models.** Release hydrographs act as upstream boundary conditions for Saint-Venant routing (Chapter 7); satellite-observed inundation downstream provides an independent check on the routed result.
4. **Cloud cover is the practical limitation of optical imagery.** The reservoir and catchment were under heavy monsoon cloud during the visit (Figure 8.2). Optical satellites such as Sentinel-2 cannot see the ground in these conditions, precisely when floods occur, which reinforces the plan to add cloud-penetrating Sentinel-1 SAR.
5. **Fast hilly catchments leave little lead time.** The steep Sahyadri catchment around the reservoir responds quickly to heavy rain, so warning times are short, consistent with the sensitivity analysis showing that larger floods travel faster (Table 7.2). Rapid, automated mapping is therefore more valuable than slow manual mapping.

8.4 Industry visit (proposed)

A visit to ClimMaTech Pvt. Ltd. (IIT Mandi, Kamand Campus, District Mandi, Himachal Pradesh – 175075) is proposed, with dates to be fixed in consultation with the industry mentor, to review the hydrological platform with the product team, demonstrate Prithvi Flood-IN, and agree on the integration of flood maps and the Mahanadi pilot.

9 Discussion and Limitations

9.1 Key findings so far

1. Fine-tuning a global flood foundation model on 100 Indian events measurably improves it on unseen Indian floods (IoU +11%, precision +43%), primarily by suppressing false alarms.
2. The improvement comes with reduced recall; the model's operating point must be chosen for the application, and threshold calibration is the cheapest lever.
3. In the controlled benchmark, a multimodal foundation model outperformed a from-scratch U-Net by about 0.04 IoU, but the U-Net was strong and label-efficient, so a U-Net baseline must be included for the Indian dataset before drawing firm conclusions.
4. Channel roughness and slope, the uncertain inputs of the existing platform, strongly influence routed peaks, which supports the value of observed flood extents for model constraint.

9.2 Limitations and mitigations

- **Small dataset (100 events):** limited diversity of regions, seasons and land cover; about 2,680 further NDEM layers remain to be screened.
- **Optical-only input:** cloudy monsoon scenes are unusable and many NDEM maps were discarded; Sentinel-1 SAR fusion is planned.
- **Reduced recall (0.71 $\rightarrow$ 0.53):** more missed flood area; to be addressed by threshold calibration and class-weighted or Dice loss.
- **Label quality:** NDEM polygons are generalised, so boundary errors penalise IoU; boundary-tolerant metrics and visual audits are planned.
- **No from-scratch baseline on Indian data and a single training run:** the value of pre-training and run-to-run variance are not yet quantified; a U-Net baseline and repeated seeds are planned.
- **Free-tier compute and a synthetic routing experiment:** stable GPU access will be sought from the mentors, and the routing solver will be validated on PINN cases and gauge data.

10 Work Plan for the Second Half

10.1 Milestones

Table 10.1: Project milestones and status as of 5 October 2026.

No.	Milestone	Target	Status
M1	Literature review; platform and Prithvi-EO-2.0 study	Aug 2026	Completed
M2	Saint-Venant study, routing solver and PINN codebase study	Aug–Sep 2026	Completed
M3	Indian flood dataset (100 events, 3,584 chips)	Sep 2026	Completed
M4	Baseline evaluation and full fine-tuning	Sep 2026	Completed
M5	Benchmark study (Sen1Floods11)	Sep 2026	Completed
M6	Error analysis, threshold calibration, U-Net baseline, FastFlood	Oct 2026	Planned
M7	Dataset expansion and Sentinel-1 SAR fusion; PINN setup, Phase 1, training and validation	Oct–Nov 2026	Planned
M8	Mahanadi pilot and dashboard prototype	Nov 2026	Planned
M9	Final report, code, model and presentation	End of semester	Planned

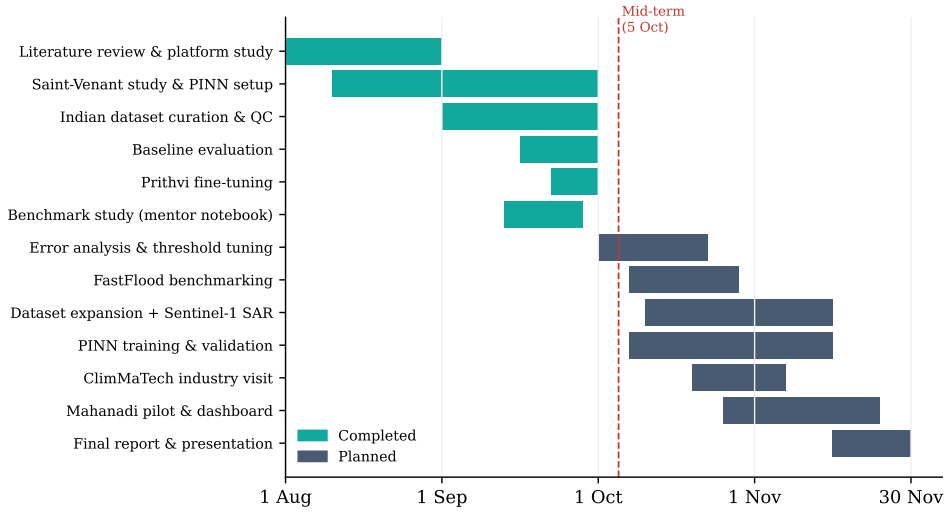


Figure 10.1: Project timeline (Gantt chart).

10.2 Planned technical work

1. **Error analysis:** per-event metrics, confusion matrices and error maps to identify failure modes (turbid water, urban areas, cloud edges, flooded vegetation).

2. **Threshold calibration:** sweep the flood-probability threshold on the validation set and report precision–recall and F1 curves; choose application-specific operating points.

3. **Loss design:** evaluate class-weighted cross-entropy and Dice loss to recover recall.

4. **Baselines:** train a U-Net from scratch on the Indian split under the same protocol, so that the benefit of pre-training is isolated.

5. **Data expansion and SAR fusion:** screen the remaining NDEM layers; add Sentinel-1 VV/VH to

recover cloudy events.

6. **External benchmarking:** compare against FastFlood validation extents where available.
7. **PINN track:** verify the PINN setup, complete Phase-1 data exploration, train and validate on the assigned river reaches, and compare with the finite-difference solver.
8. **Mahanadi pilot:** apply the model to recent Mahanadi events and overlay with Hirakud releases and CWC gauges in a dashboard prototype.

10.3 Expected deliverables

The final deliverables are: the India-specific flood segmentation dataset with its documented curation procedure; the fine-tuned Prithvi Flood-IN checkpoint with training and evaluation code; an evaluation report with per-event metrics and error maps; an inference pipeline producing georeferenced flood-extent maps; a dashboard prototype for the Mahanadi basin; the Saint-Venant and PINN study; and the final report and presentation.

10.4 Roles and responsibilities

All members share common work: understanding Prithvi-EO-2.0 and its training methodology, participating in reviews, and contributing to reports and presentations.

Table 10.2: Primary responsibilities of team members.

Member	Primary responsibilities
Nikhil Upadhyay	Research on applying the fine-tuned model for flood prediction and analysis; inference use cases, dashboard design and Mahanadi pilot; reports and presentations
Prakhar Gupta	Data used to train Prithvi; Indian fine-tuning dataset: scene–label pairing, masking, event-wise split, storage optimisation
Prabhat Kumar	Applications of the fine-tuned model; Saint-Venant study, routing experiments and PINN code base; presentations
Vineet Jangir	Training methodology of foundation and fine-tuned models; execution of model training, environment and checkpoint management
Himanshu Shete	Performance evaluation (IoU, precision, recall, F1, confusion matrix, inference time); baseline comparison and error maps; inference workflow

11 Conclusions

In the first half of the project the team has built an India-specific flood segmentation dataset from official NDEM maps and Sentinel-2 imagery under a strict event-wise protocol, fully fine-tuned the 300 M-parameter Prithvi-EO-2.0 flood model on free-tier infrastructure, and shown on 16 unseen flood events that fine-tuning improves IoU by 11% and precision by 43%, at the cost of lower recall. A mentor-provided benchmark confirmed that a multimodal foundation model can outperform strong from-scratch baselines, while also showing how strong and label-efficient a U-Net can be. On the physics side, the team implemented and analysed a Saint-Venant routing solver, quantified the sensitivity of flood peaks to uncertain channel parameters, and set up a PINN framework. Together with the inference workflow, the application analysis and the Pawna Dam field visit, this establishes the foundation for the second half: calibrating

the model's operating point, adding radar data, validating against independent benchmarks and piloting an integrated flood information product for the Mahanadi basin.

References

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